

DATA LINK LAYER

Module 2

Error Detection and Correction using Hamming Code:

- Hamming code is an error-correcting code used to detect and correct single-bit errors in transmitted data.
- The basic idea is to add few extra parity bits to the original data bits.
- These parity bits allow the receiver to determine which bit is erroneous and correct it.

Step followed by the sender:

- 1. Number of parity bits:** If there are m data bits and r parity bits, the number of parity bits must satisfy: $2^r \geq m + r + 1$, where: m = number of data bits and r = number of parity bits.
- 2. Position of parity bits:** Parity bits are placed at positions that are powers of 2: 1, 2, 4, 8, 16,...

For example, with 4 data bits and 3 parity bits, the positions are:

Position	1	2	3	4	5	6	7
Bit	P_1	P_2	D_1	P_4	D_2	D_3	D_4

So the transmitted 7-bit Hamming code (Codeword) is: $P_1P_2D_1P_4D_2D_3D_4$
Parity bits are unknown. To calculate it goto the 3rd step.

3. How parity bits work:

Each parity bit checks specific positions:

P_1 checks positions 1, 3, 5, 7 (write one skip one, start from 1): $P_1 D_1 D_2 D_4$

P_2 checks positions 2, 3, 6, 7 (write two skip two, start from 2): $P_2 D_1 D_3 D_4$

P_4 checks positions 4, 5, 6, 7 (write four skip four, start from 4): $P_4 D_2 D_3 D_4$

The parity bits are calculated so that each group has the required parity, usually even parity.

Steps followed by the receiver:

1. Error detection and correction:

At the receiver, the parity checks are performed again.

Suppose the parity-check results are:

S_1 checks positions 1, 3, 5, 7 for parity bit P_1

S_2 checks positions 2, 3, 6, 7 for parity bit P_2

S_4 checks positions 4, 5, 6, 7 for parity bit P_4

These syndrome bits are interpreted as:

$$(S_4 S_2 S_1)_{10}$$

Therefore, bit $(S_4 S_2 S_1)_{10}$ is erroneous.

The receiver corrects the error by flipping bit $(S_4 S_2 S_1)_{10}$.

Example 1: Hamming (7, 4)

Dataword: 1011

Codeword: 0110011 (obtained after adding parity bits)

Sent: 0110011

Received: 0110111

Syndrome: $(101)_2$ i.e. bit 5 is corrupted. Flip it to get the correct codeword.

Correct Codeword: 0110011

Example 2: Solve it.

Dataword: 10011010

Hamming Code vs Parity Bit:

Feature	Simple Parity	Hamming Code
Purpose	Mainly error detection	Error detection and correction
Extra Bits	Usually one parity bit	Multiple parity bits
Locate Error	Cannot normally locate exact erroneous bit	Can locate a single erroneous bit
Correct Error	No	Yes, for a single-bit error
Complexity	Simple	More complex



THANK YOU